

# AI511/MM505 Linear Algebra with Applications

## Lecture 10 – Rank and Nullity, Determinants

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# Schedule

| Week      | Lectures      | Exercise sessions    |
|-----------|---------------|----------------------|
| 36        | Theory        | No exercise sessions |
| 37        | Theory        | Theory               |
| 38        | Applications  | Applications         |
| 39        | Theory        | Theory               |
| 40        | Theory        | Theory               |
| <b>41</b> | <b>Theory</b> | <b>Theory</b>        |
| 42        | Autumn break  |                      |
| 43        | Theory        | Theory               |
| 44        | Applications  | Theory               |

- Week 44 is split: lectures cover applications, exercises cover theory.

# Outline

Recap

Rank and nullity of a matrix

Determinants

## Recap

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# Coordinates

- Suppose that  $B = \{v_1, \dots, v_k\}$  is a basis of a subspace  $\mathcal{S} \subset \mathbb{R}^n$ . Then  $c_1, \dots, c_n \in \mathbb{R}$  such that

$$v = c_1 v_1 + \dots + c_k v_k$$

are called the coordinates of  $v \in \mathcal{S}$  and  $c_1, \dots, c_k \in \mathbb{R}$  are uniquely determined by  $v \in \mathcal{S}$ .

- We write

$$[v]_B = (c_1, \dots, c_n)$$

to indicate that  $c_1, \dots, c_k \in \mathbb{R}$  are the coordinates of  $v \in \mathcal{S}$  in basis  $B$ .

# Dimension

- The number of elements in some/any basis is called the dimension of a subspace  $\mathcal{S} \subset \mathbb{R}^n$  and denoted by  $\dim(\mathcal{S})$ .
- We formally define the dimension of  $\{0\}$  to be 0.
- $\dim(\mathbb{R}^n) = n$ .
- $\dim(\text{span}\{v_1, v_2\}) = 2$ , where  $v_1, v_2 \in \mathbb{R}^n$  with  $n \geq 2$  are two linearly independent vectors.

# Bases and dimension

- Suppose that  $\mathcal{S} \subset \mathbb{R}^n$  is a subspace and  $\dim(\mathcal{S}) = k$ . If  $|B| = k$ , then  $B$  is a basis if and only if  $\text{span}(B) = \mathcal{S}$  or if  $B$  is linearly independent.
- In simple terms, if we know the dimension of the space, it is enough to check if the vectors span the space or if the vectors are linearly independent to determine if the vectors form a basis.

# Change of basis

## Definition 3.1.2 of Johnston (2021)

Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^n$  with bases  $B = \{v_1, v_2, \dots, v_k\}$  and  $C$ . The *change-of-basis matrix* from  $B$  to  $C$ , denoted by  $P_{C \leftarrow B}$ , is the  $k \times k$  matrix whose columns are the coordinate vectors  $[v_1]_C, [v_2]_C, \dots, [v_k]_C$ :

$$P_{C \leftarrow B} := \begin{bmatrix} [v_1]_C & [v_2]_C & \dots & [v_k]_C \end{bmatrix}.$$

## Theorem 3.1.3 of Johnston (2021)

Suppose  $B$  and  $C$  are bases of a subspace  $\mathcal{S}$  of  $\mathbb{R}^n$ , and let  $P_{C \leftarrow B}$  be the change-of-basis matrix from  $B$  to  $C$ . Then

- (a)  $P_{C \leftarrow B}[v]_B = [v]_C$  for all  $v \in \mathcal{S}$ , and
- (b)  $P_{C \leftarrow B}$  is invertible and  $P_{C \leftarrow B}^{-1} = P_{B \leftarrow C}$ .

Furthermore,  $P_{C \leftarrow B}$  is the unique matrix with property (a).

## Rank and nullity of a matrix

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# Inverse of a matrix revisited

## Theorem 2.2.4 of Johnston (2021)

Suppose that  $A \in \mathbb{R}^{n \times n}$ . The following are euquivalent:

- (a)  $A$  is invertible.
- (b) The system of linear equations  $Ax = b$  has a solution for all  $b \in \mathbb{R}^n$ .
- (c) The system of linear equations  $Ax = 0$  has a unique solution.

- Hence, the range of an invertible matrix is all of  $\mathbb{R}^n$  (think why this is the case).
- Geometrically speaking, invertible matrices do not “squash” space down into a smaller-dimensional subspace.

# Non-invertible matrices

- If  $A \in \mathbb{R}^{n \times n}$  is non-invertible, it means that there is at least one  $b \in \mathbb{R}^n$  such that  $Ax \neq b$  for any choice of  $x \in \mathbb{R}^n$ .
- It follows that  $\text{range}(A)$  is strictly smaller than  $\mathbb{R}^n$ , i.e., there is at least one  $b \in \mathbb{R}^n$  such that  $b \notin \text{range}(A)$ .
- The range of some non-invertible matrices is smaller while the range of some other non-invertible matrices is bigger.
- The size of  $\text{range}(A)$  can be used to describe how non-invertible the matrix is (the smaller the range of a matrix, the further the matrix is away from being invertible).
- How can we describe the size of  $\text{range}(A)$ ?

## Definition 2.4.4 of Johnston (2021)

The *rank* of  $A \in \mathbb{R}^{m \times n}$  is denoted by  $\text{rank}(A)$  and defined by

$$\text{rank}(A) = \dim(\text{range}(A)).$$

- Observe that the rank is defined for all matrices (not necessarily square) but our intuition is based on invertibility of matrices which is only defined for square matrices.
- Recommended to watch *Inverse matrices, column space, and null space* of 3Blue1Brown.

# Nullity of a matrix

Somewhat complementary to the rank of a matrix is its nullity, which is the dimension of its null space.

## Definition

The *nullity* of  $A \in \mathbb{R}^{m \times n}$  is denoted by  $\text{nullity}(A)$  and defined by

$$\text{nullity}(A) = \dim(\text{null}(A)).$$

$A \in \mathbb{R}^{n \times n}$  is invertible if and only if the system of linear equations  $Ax = 0$  has a unique solution (i.e., its null space is  $\{0\}$ ), which is equivalent to it having nullity 0.

## Theorem 2.4.9 of Johnston (2021)

Let  $A \in \mathbb{R}^{n \times n}$ . The following are equivalent:

- (a)  $A$  is invertible.
- (b)  $\text{rank}(A) = n$ .
- (c)  $\text{nullity}(A) = 0$ .

## Relationship between $\text{rank}(A)$ and $\text{nullity}(A)$

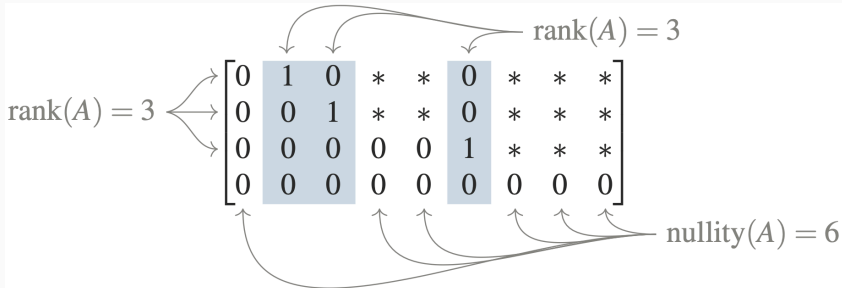
- Recall that multiplying  $A$  by the vector  $x$  is equivalent to forming a linear combination of the columns of  $A$ , using the entries of  $x$  as the weights.
- If there are more ways to linearly combine the columns of  $A$  into the  $0$  vector,  $\text{null}(A)$  and  $\text{nullity}(A)$  are larger.
- Intuitively, the more ways you can combine the columns of  $A$  to get zero, the fewer ways there should be to combine them to reach any other non-zero vector.
- There should be a relationship between  $\text{rank}(A)$  and  $\text{nullity}(A)$ . How does it look like?

## Theorem 2.4.10 of Johnston (2021)

Suppose that  $A \in \mathbb{R}^{m \times n}$ . Then

$$\text{rank}(A) + \text{nullity}(A) = n.$$

## Rank-nullity theorem (cont.)



- The rank of a matrix equals the number of leading columns that its reduced REF has, which equals the number of non-zero rows.
- Its nullity is the number of non-leading columns in its reduced REF, which equals the number of free variables.

Figure 2.24 of Johnston (2021)

- We have that

$$\text{rank}(A) \leq \min\{m, n\}$$

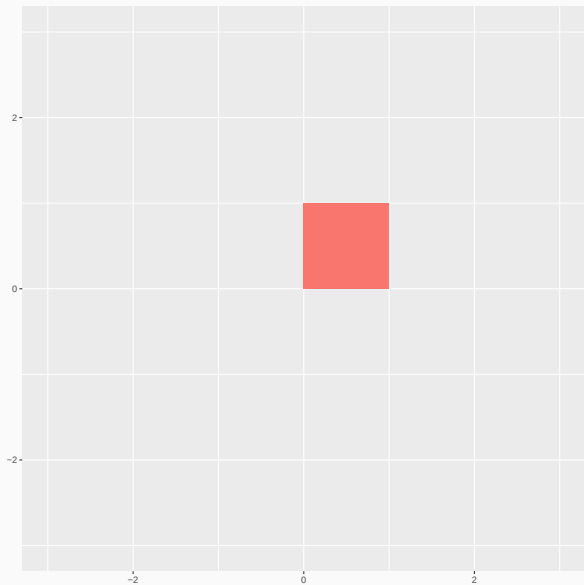
for  $A \in \mathbb{R}^{m \times n}$  (think why this inequality is true).

- We say that  $A$  has *full rank* if  $\text{rank}(A) = \min\{m, n\}$ .

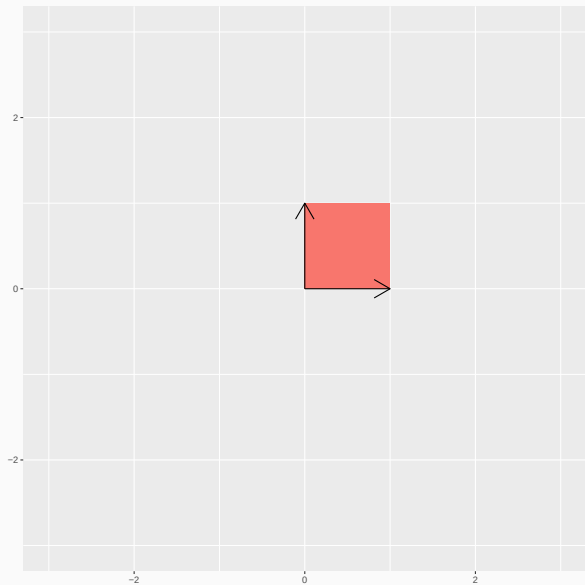
# Determinants

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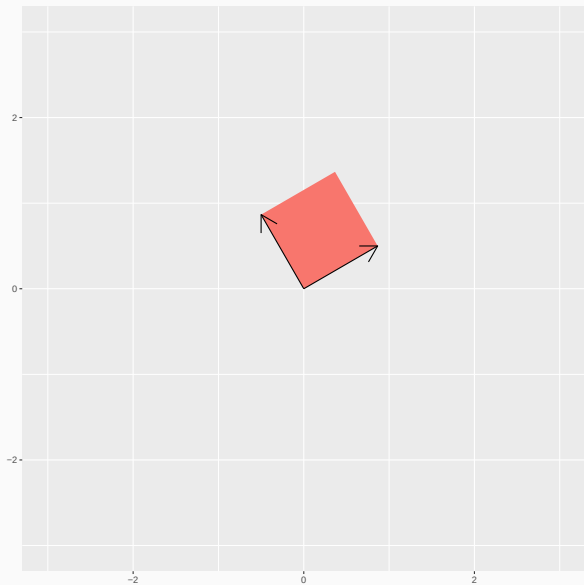
# Unit square



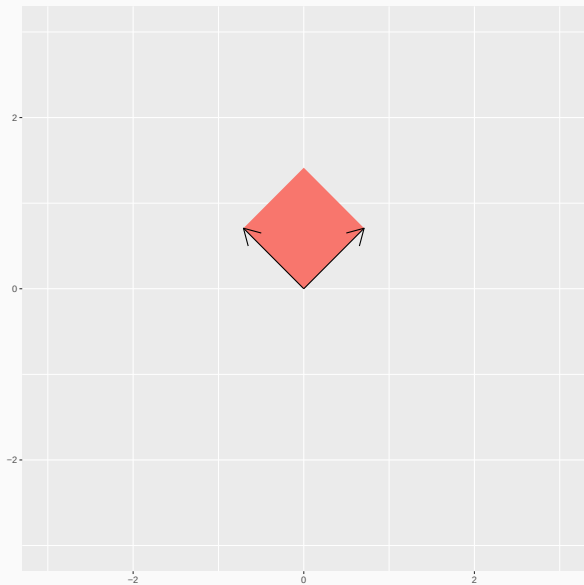
# Unit square



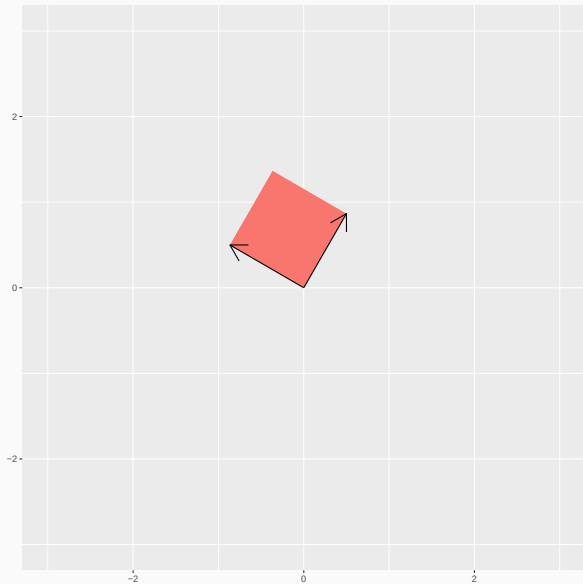
# Rotation by $\pi/6$ radians



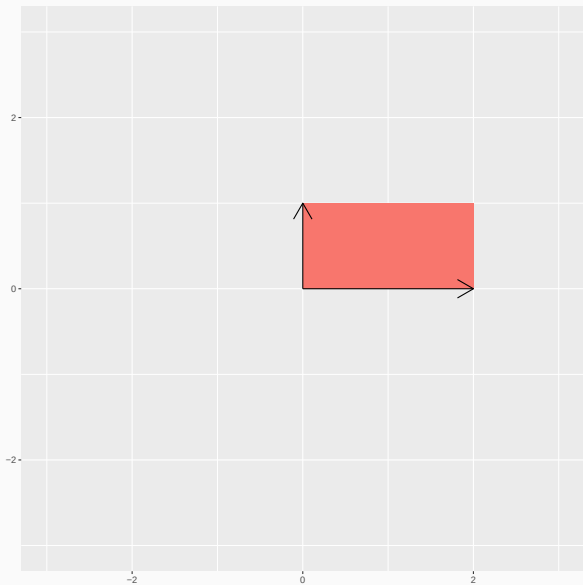
# Rotation by $\pi/4$ radians



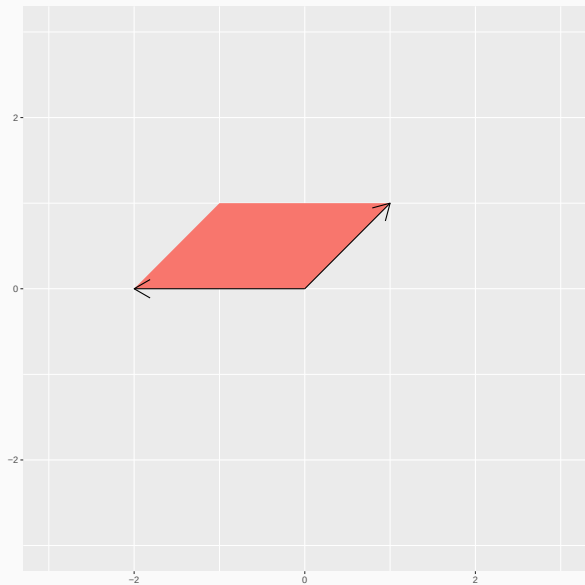
# Rotation by $\pi/3$ radians



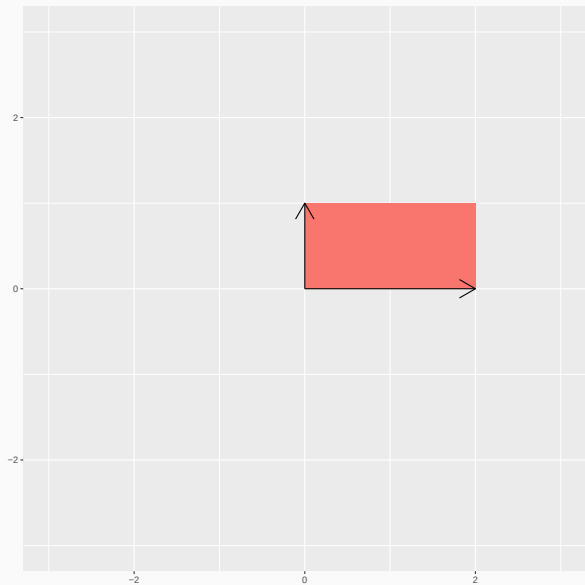
## Stretching horizontally by 2



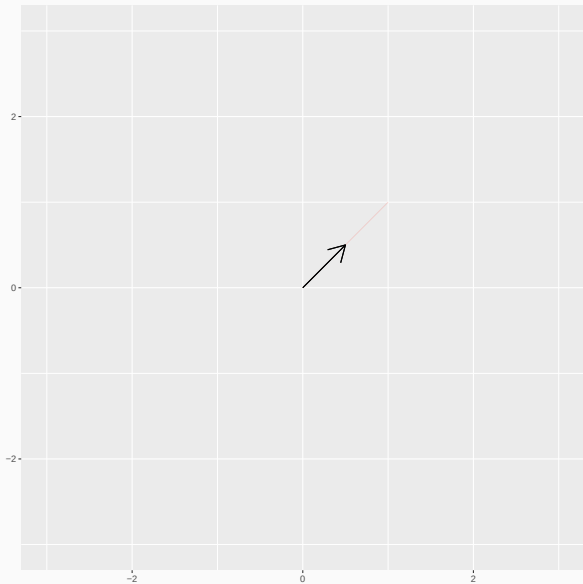
# Matrix $B$



# Matrix A



# Projection onto the $y = x$ line



# Determinant

- The determinant of  $A$ , which we denote by  $\det(A)$  or  $|A|$ , is the area (or volume, or hypervolume, depending on the dimension  $n$ ) of this distortion of the unit square.
- In other words, it measures how much  $A$  expands space when acting as a linear transformation—it is the ratio

$$\frac{\text{volume of output region}}{\text{volume of input region}}.$$

- Recommended to watch *The determinant* of 3Blue1Brown.

# Definition

## Definition 3.2.1 of Johnston (2021)

The determinant is the unique function  $\det : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$  that satisfies the following three properties:

- (a)  $\det(I_n) = 1$ ;
- (b)  $\det(AB) = \det(A) \det(B)$ ;
- (c) for all  $c \in \mathbb{R}$  and all  $v, w, a_1, a_2, \dots, a_n \in \mathbb{R}^n$ , it is the case that

$$\det([a_1 \mid \cdots \mid v + cw \mid \cdots \mid a_n]) = \det([a_1 \mid \cdots \mid v \mid \cdots \mid a_n]) + c \cdot \det([a_1 \mid \cdots \mid w \mid \cdots \mid a_n]).$$

Property (c) is sometimes called multilinearity of the determinant, which in this context just means “linear in each of the columns”.

# Properties of the determinant

## Theorem 3.2.1 of Johnston (2021)

Suppose  $A \in \mathbb{R}^{n \times n}$ . Then  $A$  is invertible if and only if  $\det(A) \neq 0$ , and if it is invertible, then  $\det(A^{-1}) = 1/\det(A)$ .

## Theorem 3.2.2 of Johnston (2021)

Suppose that  $A \in \mathbb{R}^{n \times n}$  and  $c \in \mathbb{R}$ . Then

- (a)  $\det(cA) = c^n \det(A)$ ;
- (b)  $\det(A^T) = \det(A)$ .

# Computation

- There are several different formulas for computing  $\det(A)$ , and they are all somewhat ugly and involved.
- The determinant of every non-invertible matrix is 0 so we need a method of computing the determinant of invertible matrices.
- Recall that  $A$  is invertible if and only if

$$E_k \dots E_1 A = I_n,$$

where  $E_k, \dots, E_1$  are elementary matrices.

- Hence,

$$\det(A) = \frac{1}{\det(E_k) \det(E_{k-1}) \dots \det(E_1)}.$$

- How do we calculate determinants of elementary matrices?

# Determinants of elementary matrices

- The row operation  $cR_i$  has determinant  $c$ .
- The row operation  $R_i + cR_j$  has determinant 1.
- The row operation  $R_i \leftrightarrow R_j$  has determinant  $-1$ .

## Determinant as signed area

- A linear transformation maps the unit square to a parallelogram.
- The absolute value of the determinant gives the area of this parallelogram

$$|\det(A)| = \text{area}.$$

- The sign of the determinant tells us whether the transformation preserves or reverses orientation of the coordinate directions.
- If  $\det(A) > 0$ , the coordinate directions keep their orientation.
- If  $\det(A) < 0$ , the coordinate directions are flipped.

# Computing determinants via Gauss-Jordan elimination

## Theorem 3.2.3 of Johnston (2021)

Let  $A \in \mathbb{R}^{n \times n}$  be invertible, and suppose that  $A$  can be row-reduced to  $I_n$  via  $k$  “multiplication” row operations  $c_1 R_{i_1}, c_2 R_{i_2}, \dots, c_k R_{i_k}$  as well as  $s$  “swap” row operations (and some “addition” row operations that we do not care about). Then

$$\det(A) = \frac{(-1)^s}{c_1 c_2 \dots c_k}.$$

Are there any types of matrices for which calculating the determinant is straightforward?

# Diagonal matrices

## Definition

$A \in \mathbb{R}^{n \times n}$  is said to be *diagonal* if  $a_{ij} = 0$  for  $i \neq j$ .

## Examples

$$\begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

If  $A \in \mathbb{R}^{n \times n}$  is diagonal, we also write

$$A = \text{diag}(a_1 \ \dots \ a_n).$$

## Diagonal matrices (continued)

Matrix multiplication of diagonal matrices is performed elementwise

$$\begin{aligned}\text{diag}(a_1 \ \dots \ a_n) \text{diag}(b_1 \ \dots \ b_n) &= \\ &= \text{diag}(a_1 b_1 \ \dots \ a_n b_n).\end{aligned}$$

### Theorem

*A diagonal matrix  $D \in \mathbb{R}^{n \times n}$  is invertible if and only if  $d_i \neq 0$  for all  $i = 1, \dots, n$  and in which case  $D^{-1} = \text{diag}(d_1^{-1} \ \dots \ d_n^{-1})$ .*

It is straightforward to calculate powers of a diagonal matrix  $D$  since

$$D^k = \text{diag}(d_1^k \ \dots \ d_n^k)$$

for  $k \geq 1$ .

# Triangular matrices

## Definition

$A \in \mathbb{R}^{n \times n}$  is called *upper (lower) triangular* matrix if all entries below (above) the main diagonal are zero, i.e.,  $a_{ij} = 0$  for  $i > j$  ( $a_{ij} = 0$  for  $i < j$ ).

## Examples

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}.$$

Upper triangular                      Lower triangular

So diagonal means upper *and* lower triangular.

# Determinants of triangular matrices

## Theorem 3.2.4 of Johnston (2021)

Suppose  $A \in \mathbb{R}^{n \times n}$  is a triangular matrix. Then  $\det(A)$  is the product of its diagonal entries:

$$\det(A) = a_{11} \cdot \dots \cdot a_{nn}.$$

Example

$$\begin{vmatrix} 1 & 2 & -1 & 7 & 91 \\ 0 & 1 & 19 & 0 & \frac{1}{10} \\ 0 & 0 & 2 & 1 & -5 \\ 0 & 0 & 0 & 1 & \pi \\ 0 & 0 & 0 & 0 & 3 \end{vmatrix} = 1 \cdot 1 \cdot 2 \cdot 1 \cdot 3 = 6.$$

Recall that a diagonal matrix is also triangular.

## Determinant of $2 \times 2$ matrices

### Theorem 3.2.6 of Johnston (2021)

The determinant of a  $2 \times 2$  matrix is given by

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc.$$

We saw this expression when we talked about the inverse of a  $2 \times 2$  matrix and we saw that a  $2 \times 2$  matrix is invertible if and only if  $ad - bc \neq 0$ .

## Definition 3.2.2 of Johnston (2021)

Suppose  $A \in \mathbb{R}^{n \times n}$  and let  $1 \leq i, j \leq n$ . Then

- (a) the  $(i, j)$ -minor of  $A$  is the determinant of the  $(n - 1) \times (n - 1)$  matrix that is obtained by deleting the  $i$ -th row and  $j$ -th column of  $A$ , and
- (b) the  $(i, j)$ -cofactor of  $A$  is the quantity  $(-1)^{i+j} m_{ij}$ , where  $m_{ij}$  is the  $(i, j)$ -minor of  $A$ .

# Computing determinants via cofactor expansion

The determinants of large matrices can be expressed in terms of determinants of smaller matrices.

## Theorem 3.2.8 of Johnston (2021)

Suppose  $A \in \mathbb{R}^{n \times n}$  has cofactors  $\{c_{ij}\}_{i,j=1}^n$ . Then

$$\det(A) = a_{i1}c_{i1} + a_{i2}c_{i2} + \dots + a_{in}c_{in} \quad (1)$$

$$\det(A) = a_{1j}c_{1j} + a_{2j}c_{2j} + \dots + a_{nj}c_{nj} \quad (2)$$

for all  $i, j = 1, \dots, n$ .

- (1) is called a cofactor expansion along the  $i$ -th row.
- (2) is called a cofactor expansion along the  $j$ -th column.

# Examples

- We have that

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a(-1)^{1+1}d + b(-1)^{1+2}c = ad - bc.$$

- Suppose that

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}.$$

Show that  $\det(A) = -1$ .

## Tips and tricks

- Instead of row operations, one can do column operations. This corresponds to row operations on  $A^T$ .
- If a matrix  $A$  has a row or column with a lot of zeros, then this is often good for row/column cofactor expansion.
- If a matrix  $A$  has a zero row (or column) then  $\det(A) = 0$ .
- If two rows (columns) are proportional, then  $\det(A) = 0$ , since one can then create a zero-row (zero-column) by row operations.



Johnston, Nathaniel (2021). *Introduction to linear and matrix algebra*. Springer Cham.